

A Far Field Attractor in the Umbilicus Score of Volume Cartographer

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Abstract—The umbilicus estimator of Volume Cartographer scores a candidate centre with a weighted sum of squared cosines divided by the sum of the weights. That quotient is a weighted mean, whose far field limit, the largest eigenvalue of the normals’ covariance, can exceed its score on the axis; where it does, the climb walks out of the volume, as on 134 of the 135 slices of 360 where the published estimate leaves the grid, measured on the shipped function. Removing the division makes it a weighted sum, which must have a maximum at finite distance: one line. Robustness stands on its own, 576 of 576 slices inside the grid on 24 scrolls at 1 seed each, against 350 as published. Accuracy is bounded by the reference: the sum beats the centroid by the pre-registered 0.30 mm on 9 of 10 third party umbilici, out of sample, and on 4 of 5 of the challenge’s, while the error of the references themselves stays unmeasured: no published scroll carries two hand annotations, and the second umbilicus of PHerc. 1218 is a centroid of the papyrus mask and not a second hand. The cost is a measured bias towards the denser side, 0.44 to 2.44 mm on synthetic sections whose centre is known by construction, already inside the distances above rather than added to them. The claim is that attractor and its removal, not that the weighted sum is the right objective; the change offered is the line.

Index Terms—Herculaneum papyri, virtual unwrapping, umbilicus, objective functions, software defects, pre-registration.

I. The line

SPIRAL fitting [1] reconstructs the geometry of a rolled papyrus by fitting the whole spiral at once, and it has one input that cannot be omitted: the umbilicus, the polyline that follows the centre of the winding through the volume. That input is made by hand [2], the challenge publishes one for a small minority of its scrolls, and on the rest the axis has to be estimated before anything else can be attempted.

volume-cartographer [3], the toolchain of the villa repository, contains an estimator for exactly this in its core library, working over the published normal grids. It samples line segments from the grid, takes the normal of each, scores a candidate centre by how well the directions from the candidate to the segments line up with those normals, picks the best of a thousand random candidates and refines it with a hill climb. It is the function `align_and_extract_umbilicus`, defined at `normalgridtools.cpp:15`, whose header comment calls it “a RANSAC-based approach to find a common umbilicus point”. At the commit measured here nothing in the repository calls it but its own tests, and no accuracy had been reported for it. Every line number in this paper refers to commit 23adee047, and the line at issue is still there: at 2dcfaf6a0, the tip of the default branch on 2026-09-17 and 50 commits later, the divided return is still line 116, and 0 of those commits touch the source file or its header.

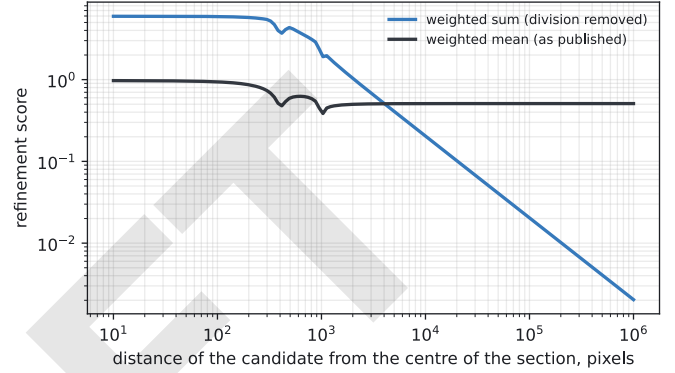


Fig. 1. Refinement score of `normalgridtools.cpp:98–117` on a synthetic section, one turn of a spiral of 4000 sampled points, against the distance of the candidate from the centre of the section. Weighted sum and weighted mean, same section, same samples. Recomputed by `tools/synthetic_density.py` on this section and on a concentric circle (`evidence/mechanism-curve.csv`): a million units away the sum retains 3.4e-4 of its maximum and the mean 0.52; on the circle, 9.5e-4 and 0.50.

The change proposed upstream is one line of that function, the return of its refinement score, with two additions beside it that need no measurement of their own: a seed argument for the generator, so that a run can be repeated, and a regression test on synthetic sections whose centre is known.

II. The defect

The refinement score runs over sampled segment midpoints, weights each by the inverse of its distance from the candidate and adds up the squared cosine between the segment normal and the direction to the segment. The last line of it divides (`normalgridtools.cpp:112–116`):

```
float weight = 1.0f / std::max(100.0f, dist);
score += (cos_angle * cos_angle) * weight;
wsum += weight;
...
return score/wsum;
```

Without that division the quantity is a weighted sum. Let D be the distance of the candidate from the section and $\mathbf{u}$ the unit vector from the candidate towards it. Every weight tends to $1/D$, so the sum tends to zero: a positive continuous function that vanishes at infinity attains a maximum at finite distance, by construction. With the division it is a weighted mean. Numerator and denominator vanish together, every direction from the candidate to a segment tends to $\mathbf{u}$, and the limit is

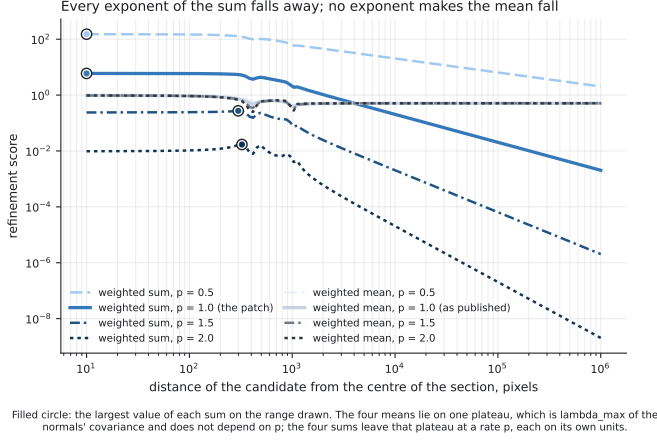


Fig. 2. The same score against the same distance as Fig. 1, with the exponent of the weight set to 0.5, 1, 1.5, 2. The four weighted means end on one plateau, the largest eigenvalue of C , which does not depend on the exponent; the four weighted sums leave it downwards with slope p . Drawn by tools/figure_exponent.py from evidence/exponent-mechanism-curve.csv; the rate at which each curve approaches or leaves the plateau is measured in evidence/exponent-ablation-cancellation-rate.csv.

the plain mean of the squared cosines over the N sampled normals $\mathbf{n}_i$,

$$\lim_{D \rightarrow \infty} \frac{\sum_i w_i \cos^2 \theta_i}{\sum_i w_i} = \frac{1}{N} \sum_{i=1}^N (\mathbf{n}_i \cdot \mathbf{u})^2 = \mathbf{u}^T C \mathbf{u}, \quad (1)$$

with $C = \frac{1}{N} \sum_i \mathbf{n}_i \mathbf{n}_i^T$. Every normal contributes equally to that limit, however far it is, because a mean does not know how many normals are contributing or where they are. On a complete circle of radial normals the limit is $1/2$ for every $\mathbf{u}$; on a partial section whose normals are coherent it rises towards 1 along the mean normal, and it can exceed the score at the true centre once the normals there are noisy, so the mean stays high far away while the sum collapses (Fig. 1).

The maximum of that limit over $\mathbf{u}$ is the largest eigenvalue of C , which turns the observation of Fig. 3 into a criterion computable from the normals alone: the defect strikes partial predictions with coherent normals and spares complete sections. Measured with the 10,000 samples the estimator draws (tools/far_field_eigenvalue.py, evidence/far-field-eigenvalue.csv), the largest eigenvalue on the PHerc0826 slice of Fig. 3 is 0.65 against a weighted mean of 0.58 at the published control point, so the far field is 1.11 times the score at the centre and the climb has somewhere better to go. On a synthetic full circle it is 0.50 against 1.00 at the centre, and on a half circle of radial normals still 0.50, because normals spanning half a turn cover both directions of the plane as evenly as normals spanning the whole of it. That is why the mean is exact on every section of Table VI, the half circle row included, where its maximum sits 0 units from the true centre against 261 for the sum. It is also why the one complete concentric section of the primary block, PHerc0332, gives the same median with and without the division, 0.35 against 0.37 mm in Table III, with 20 of 480 runs outside the grid as published against 0.

TABLE I

Equation (1) with the scroll as the unit, the 24 slices of each of the 15 scrolls with a manual reference. Flagged: slices where the far field bound exceeds the weighted mean at the reference control point. Left: slices whose estimate as published leaves the grid, by the rule of Table II. Missed: leaving runs the condition did not flag. Generated from evidence/far-field-per-scroll.csv.

scroll	flagged	left	flagged and left	missed
PHerc0125	22	22	22	0
PHerc0211	17	13	13	0
PHerc0332	1	1	1	0
PHerc0826	24	16	16	0
PHercParis4	8	5	5	0
PHerc0191	3	0	0	0
PHerc0257	9	3	2	1
PHerc0268	10	1	1	0
PHerc0358	12	10	10	0
PHerc0800	10	6	6	0
PHerc0813	8	2	2	0
PHerc1203	10	8	8	0
PHerc1218	23	23	23	0
PHerc1447	18	11	11	0
PHerc1545	17	14	14	0

TABLE II

Equation (1) against the runs, on the 360 slices of the 15 scrolls with a manual reference. Rows: whether the far field bound exceeds the weighted mean at the reference control point of that slice. Columns: whether the estimate as published leaves the grid, by the rule fixed before the run, more than half of the 20 seeds landing outside. Generated from evidence/far-field-prediction.csv.

	leaves the grid	stays inside
$\lambda_{\max} > \text{mean at the reference}$	134	58
$\lambda_{\max} \leq \text{mean at the reference}$	1	167

A faster falling weight does not rescue the mean, because a pure power cancels: with weights $1/d_i^p$ the factor D^{-p} stands in the numerator and in the denominator alike and leaves equation (1) as it is, while the sum keeps it and vanishes as D^{-p} for every $p > 0$. A larger exponent moves the distance at which the plateau becomes visible; it does not lower the plateau (Fig. 2), and Section VII measures the exponent on the fifteen scrolls.

The two objectives therefore fail on orthogonal properties. The mean fails when the normals of a section are coherent, which is what the eigenvalue measures, and it is exact when they cover the plane however one-sided the section is, as the half circle shows; the sum fails when the density of the segments is asymmetric, which is what Section VI measures, and coherence does not trouble it. A real prediction is usually both at once, partial and one-sided, so neither objective is right in general, and the choice made here is made for the sections this toolchain actually processes.

Equation (1) speaks about every slice, so it was put to every slice: on the 360 slices of the 15 scrolls that carry a manual reference, the 24 slices of each and the 20 seeds per slice that Section III measures them on. What is computed is the largest eigenvalue of C and the weighted mean at that slice's reference control point, on the samples the table's own seed draws (tools/far_field_eigenvalue.py). The reference serves only to evaluate the score at a point known to lie on the axis, which is measuring and not choosing. Table II

is the answer. The bound exceeds the score on the axis on 192 slices and the published estimate leaves the grid on 135, and 134 of those satisfy the condition: 1 of the 168 slices where the bound stays below the axis walk out, so the condition is necessary on all but 1 of the 135 slices that leave, and not sufficient, sensitivity 0.99 and specificity 0.74. The 58 slices that satisfy it and stay inside are those where the climb does not reach the border from its random start, which the criterion does not claim to predict.

The outcomes here are the shipped walk's, from the same 20 seeds per slice as the tables; the eigenvalues come from the grids and are the same under either instrument. Against the transcription's outcomes on the same slices, 3 slices change side under the majority rule: PHerc. 0257 z 8479, PHerc. 0268 z 2680, PHerc. 1203 z 5019. One of them is the miss. On PHerc. 0257 z 8479 the bound is 0.909 of the score on the axis, so the condition does not fire, and the published walk leaves the grid on 14 of 20 seeds under the shipped function against 10 of 20 under the transcription, on either side of the majority. Under the transcription the condition missed 0 slices; under the shipped function it misses 1. That is one slice at the majority line, and it is printed as a difference and not reconciled.

Those 360 slices are 24 from each of 15 scrolls, not 360 independent observations: slices of one scroll share a geometry, a surface prediction and a scanner, so a test that pools them would answer a question nobody asked. With the scroll as the unit no pooling is needed. On 14 of 15 scrolls not one run that left the grid escaped the condition; of the 14 scrolls where any run left at all, every leaving run was flagged on 13 (Table I); the remaining scroll, PHerc0191, has no run to flag. Shuffling the predictor among the slices of one scroll, independently in each scroll, holds every scroll's own two counts fixed and breaks only the pairing between them. The statistic is a sum over scrolls of independent hypergeometric counts, so its tail under that null is exact by convolution: $p = 5.29\text{e-}29$, and 0 of 200,000 shuffles reached the observed count, $p < 5.0\text{e-}6$ by simulation.

The hill climb at `normalgridtools.cpp:119-142` is therefore not searching for a maximum that is hard to find. On most real sections it is searching for one that is not there, and it keeps stepping outwards because every step improves the mean a little. Nothing stops the walk: `normalgridtools.cpp:128` adds the step to the current best with no clamp to the grid, so the answer can leave the volume by an arbitrary distance and the routine still returns it as an estimate.

Fig. 3 is one real slice scored both ways, on one scale. A weighted mean and a weighted sum do not have the same units, so both panels are divided by the score at the published control point, which is the same physical place in both. As published, the field rises towards one edge of the grid and is as high far from the sheets as it is at the control point, so its maximum sits on that edge and not on the sheets; the climb stops 42.97 mm from the control point, outside the grid. With the division removed the field collapses away from the sheets, the maximum sits inside the cloud of sheet segments, and the climb stops 5.07 mm from the control point. The hill climb behaves in both cases: it finds what its objective tells it to find.

The other score the function computes does not repair this. The candidate selection of lines 78 to 88, which picks the seed for the refinement, is an unweighted sum of the same squared cosines; it carries no distance weight at all, so it has neither decay nor weighting to localise a maximum, and on the slices where the published objective peaks at the border that one does too. Dividing by the number of samples would change nothing, while dividing by the sum of the weights moves the maximum, because that sum depends on where the candidate is.

A. Written before the confirming scrolls were looked at

The argument above is about the shape of the objective and would hold without a single scroll having been looked at. It was written down, with the criteria that would decide it, before the grids of the confirming scrolls were downloaded [4] and after 4 of the 5 scrolls of the primary block had been measured once (Section III); the run then checked it. The 10 references of the confirmation block, and every scroll of Section V without a reference, are out of sample in the strict sense that the change was not found by looking at them; the primary block is not. What the argument guarantees is narrower than what the measurement finds: a weighted sum whose weights fall as $1/D$ has a maximum at finite distance for every section, of any shape, at any resolution, but that maximum is interior to the plane and not to the grid, and it bounds the objective and not the number of sweeps the search takes to reach it. That every estimate lies inside the volume, and that every walk stopped by itself, are empirical results, 576 of 576 in Section V, and where the maximum sits is not guaranteed either: Section VI measures it.

III. Against fifteen manual references

A. What is measured

Fifteen scrolls carry a manual umbilicus: five published by the challenge, which form the primary block, and ten clicked by a third party with no stake in the result [5], which form the confirmation block. The two blocks are never pooled, and which of them is out of sample has to be said plainly. The pre-registration (prereg/preregistration-fifteen-references.md, written 2026-09-15 15:33:39Z) says in its own words that it was written before the estimator ran on any of the eleven scrolls whose grids were still to be downloaded, and before any accuracy number beyond the four already reported in the note it cites. Those four are PHerc0125, PHerc0211, PHerc0332, PHerc0826: 4 of the 5 scrolls of the primary block had been measured, with one seed, before the criteria were fixed, and 0 of the 10 of the confirmation block had. What was pre-registered is the structure, two blocks never pooled with the claim placed on the references published by the challenge, and the criteria; what was out of sample when the criteria were written is the confirmation block, whose grids were downloaded afterwards. The confirmation block therefore carries the accuracy claim, 9 of 10 below, and the primary block carries the pre-registered structure and the count of estimates leaving the volume, which is how the rest of this paper reads them. On each scroll the estimator runs on 24 slices evenly spaced along z , the estimate is a polyline through

PHerc. 0826 xy slice 8000, refinement score divided by the score at the published control point

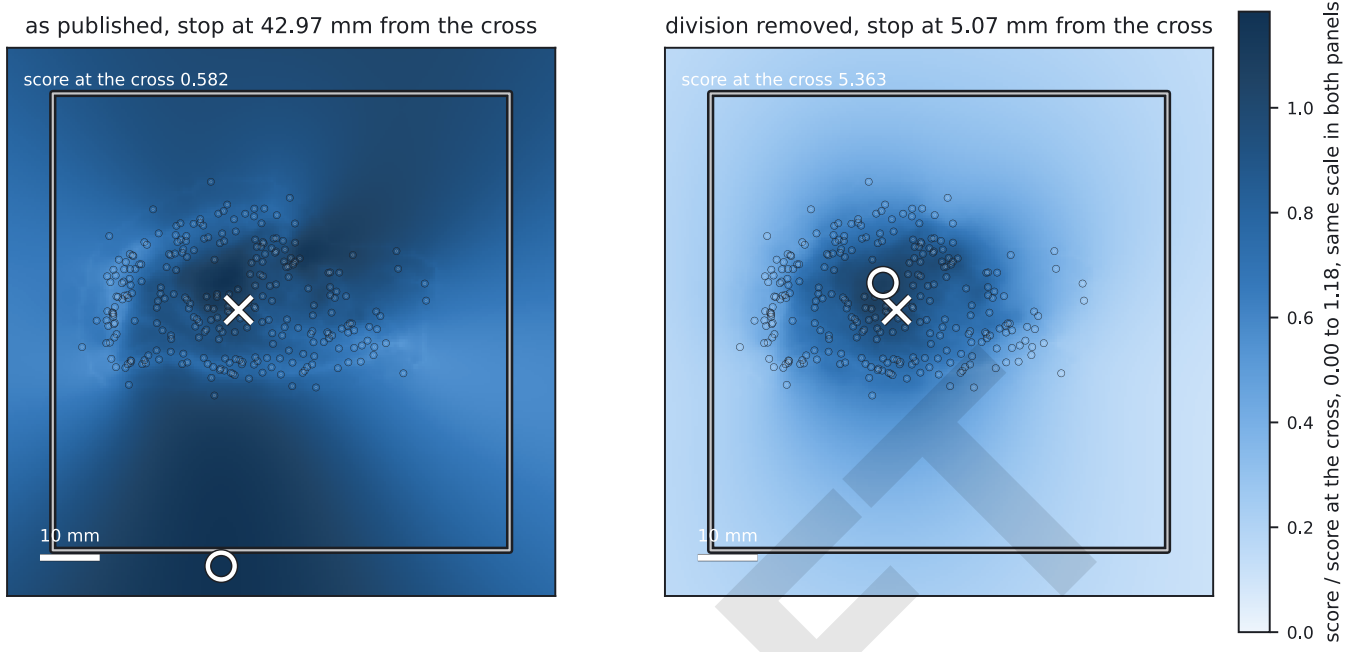


Fig. 3. Refinement score over PHerc. 0826, xy slice 8000, divided in each panel by the score at the published control point, so that both panels carry the same dimensionless scale. Left: as published, the weighted mean. Right: the weighted sum. Cross: the published control point. Circle: where the hill climb of `normalgridtools.cpp:119–142` stops. Rectangle: the bounds of the grid. Dots: sampled segment midpoints. Bar: 10 mm at 9.362 μm per voxel. Drawn by `inputs/figure_umbilicus.py` from that slice of the published normal grids and from the reference umbilicus of PHerc. 0826.

TABLE III

Primary block: the 5 umbilici published by the Vesuvius Challenge, one of them annotated by David Josey. Every rule is scored on the same 24 slices of the same scroll; the estimate is a polyline through those slices and the distance is taken at each of the scroll's control points, in millimetres of the scan the reference is annotated on, so the row is a median over 32 to 154 control points, not over slices. Bold is the estimator with the weighted sum, estimated with 20 seeds per slice: the median over the 20 per-seed medians, its 95 % percentile bootstrap interval in brackets (1,000 resamples, one seed drawn per slice and the control points resampled, generator seed 20260916). The margin is centroid minus estimator; $^{\circ}$ marks a margin whose interval includes zero, and † a margin smaller than 2.205 mm. That distance separates the hand annotation of PHerc. 1218 from a per slice centroid of the papyrus mask [6], not one hand from another, and it sits 0.06 mm from that scroll's own centroid column in Table IV, so the dagger reads as a margin smaller than the baseline's own distance from the reference, which is close to circular. It is kept because the pre-registration fixed the marking before any of these numbers were seen; Section IX gives the size of the overlap. It is a reading aid and not a criterion. The last column counts estimates outside the grid over all 480 runs, as published and with the division removed. Generated from `evidence/cpp-k20.csv`.

scroll	points	published	with the sum	[95 %]	centroid	axis	margin [95 %]	outside
PHerc0125	75	136.59	1.59	[1.16, 2.15]	2.48	1.53	+0.90 [+0.33, +1.32] †	441 to 0
PHerc0211	84	63.63	1.37	[0.89, 2.14]	2.83	8.07	+1.46 [+0.68, +1.93] †	260 to 0
PHerc0332	154	0.35	0.37	[0.28, 0.43]	0.77	1.63	+0.39 [+0.33, +0.48] †	20 to 0
PHerc0826	45	53.13	4.29	[2.90, 5.70]	3.51	7.44	-0.78 [-2.20, +0.61] $^{\circ\dagger}$	322 to 0
PHercParis4	127	4.36	1.16	[0.91, 1.52]	3.39	7.63	+2.23 [+1.87, +2.47]	103 to 0

them, and the distance is read at every control point of the reference, in millimetres of the scan the reference is annotated on; the row of Tables III and IV is the median over those 32 to 154 points. These are distances from a hand annotation, not errors: the annotation has an error of its own, and a row that is closer is closer to one person's opinion of where the axis is. The baseline is the centroid, slice by slice, of the filled sheet mask, which is the simplest rule that uses the same input; a constant axis at the centre of the field is the control that says whether a scroll discriminates at all.

The tables are computed by calling the function itself. `vc_gen_umbilicus`, the command line tool this patch adds beside the one line, reads a normal grid slice and

calls `align_and_extract_umbilicus` with the seed argument, so every row below is the compiled function and not a reading of it: 14,400 calls, the 24 slices of each scroll by 20 seeds by the two objectives, in 50 minutes on 6 cores, 5.01 core hours and 1.25 s for the mean call. Nothing bounded the search: the loop at `normalgridtools.cpp:119–142` halves its step only when none of the eight probes improves and has no iteration limit, and it was left exactly so.

That recomputation puts the claim of this paper on the shipped function rather than on a reading of it, and it is the plainest number here. Over the 7,200 estimates of each objective, on the 15 scrolls with a manual reference, the published weighted mean leaves the grid 2,727 times and

TABLE IV

Confirmation block: 10 manual references of a third party, herculaneum-umbilici [5], MIT, clicked on level 3 images at scale 8, one slice every about 480 in z, by someone who was not annotating for this purpose. A weaker reference by construction, kept in a block of its own and never pooled with Table III. Columns and marks as in Table III. Generated from evidence/cpp-k20.csv.

scroll	points	published	with the sum [95 %]	centroid	axis	margin [95 %]	outside
PHerc0191	62	6.26	4.31 [2.83, 5.71]	6.42	5.57	+2.11 [+0.71, +3.60] [†]	0 to 0
PHerc0257	40	9.34	1.73 [1.24, 2.76]	3.71	5.77	+1.97 [+0.95, +2.47] [†]	56 to 0
PHerc0268	44	9.38	4.44 [2.87, 5.83]	11.10	10.26	+6.66 [+5.27, +8.23]	30 to 0
PHerc0358	35	9.06	2.76 [1.58, 3.79]	4.90	6.26	+2.14 [+1.11, +3.32] [†]	200 to 0
PHerc0800	40	5.16	2.68 [1.77, 3.85]	6.63	9.78	+3.95 [+2.79, +4.87]	131 to 0
PHerc0813	48	10.46	7.29 [4.73, 9.26]	7.59	13.50	+0.30 [-1.67, +2.86] ^{o†}	40 to 0
PHerc1203	36	5.11	1.48 [1.00, 1.88]	4.38	4.48	+2.91 [+2.50, +3.38]	150 to 0
PHerc1218	36	101.98	2.39 [1.62, 3.63]	2.15	5.01	-0.24 [-1.48, +0.53] ^{o†}	459 to 0
PHerc1447	36	39.81	1.85 [1.23, 2.79]	3.18	6.73	+1.33 [+0.38, +1.95] [†]	235 to 0
PHerc1545	32	49.49	1.87 [1.33, 3.25]	2.40	7.05	+0.53 [-0.85, +1.07] ^{o†}	280 to 0

TABLE V

Two more rules, scored after the fact. The plateau point nearest the centroid and the maximum of the distance transform of the filled section, both from the umbilicus bench [7], on the same 24 slices, the same heights and the same control points as Tables III and IV and with the same metric; the first two columns repeat those tables so that a row reads across. Medians in millimetres with a 95 % percentile bootstrap interval over the control points (1,000 resamples, generator seed 20260916); the two rules are deterministic and take no seed. The margin is the best of the three simple rules on that scroll minus the estimator, positive where the estimator is the closer. ^t marks the 3 scrolls whose data set the plateau rule's threshold, where it is fitted and not tested. The blocks are those of Tables III and IV and are not pooled. **Both rules are post hoc with respect to every pre-registration of this paper.** declared so on 18 September 2026 and written after the frozen tables were known: the pre-registered baseline criterion names the centroid, it was fixed before the run, and nothing in this table changes a pre-registered verdict. Generated from evidence/baselines-k20.csv.

scroll	with the sum [95 %]	centroid	plateau nearest centroid [95 %]	maximum of the distance transform [95 %]	margin
<i>Primary block</i>					
PHerc0125	1.59 [1.16, 2.15]	2.48	2.06 [1.71, 2.43] ^t	2.28 [1.88, 2.76]	+0.47
PHerc0211	1.37 [0.89, 2.14]	2.83	2.85 [1.84, 4.50] ^t	4.41 [3.16, 5.04]	+1.46
PHerc0332	0.37 [0.28, 0.43]	0.77	0.74 [0.61, 0.88]	1.07 [0.91, 1.25]	+0.37
PHerc0826	4.29 [2.90, 5.70]	3.51	2.78 [1.69, 4.41] ^t	4.61 [3.75, 5.33]	-1.51
PHercParis4	1.16 [0.91, 1.52]	3.39	3.29 [2.76, 3.87]	5.05 [4.46, 5.51]	+2.13
<i>Confirmation block</i>					
PHerc0191	4.31 [2.83, 5.71]	6.42	6.41 [5.65, 7.16]	6.63 [5.50, 7.29]	+2.10
PHerc0257	1.73 [1.24, 2.76]	3.71	3.58 [2.62, 4.19]	2.27 [1.77, 3.33]	+0.54
PHerc0268	4.44 [2.87, 5.83]	11.10	11.64 [10.06, 12.93]	12.58 [11.27, 13.60]	+6.66
PHerc0358	2.76 [1.58, 3.79]	4.90	5.22 [4.44, 7.27]	6.63 [5.27, 7.67]	+2.14
PHerc0800	2.68 [1.77, 3.85]	6.63	6.39 [4.90, 7.32]	5.86 [5.05, 6.32]	+3.18
PHerc0813	7.29 [4.73, 9.26]	7.59	7.59 [6.40, 9.05]	7.70 [6.38, 8.45]	+0.30
PHerc1203	1.48 [1.00, 1.88]	4.38	4.25 [2.36, 5.24]	3.13 [2.44, 4.11]	+1.65
PHerc1218	2.39 [1.62, 3.63]	2.15	2.24 [1.59, 3.09]	4.15 [3.23, 5.67]	-0.24
PHerc1447	1.85 [1.23, 2.79]	3.18	3.12 [2.28, 4.22]	3.52 [2.50, 5.78]	+1.27
PHerc1545	1.87 [1.33, 3.25]	2.40	1.88 [1.47, 2.71]	3.98 [2.33, 5.00]	+0.01

the weighted sum 0: inside on 4,473 of 7,200 against 7,200 of 7,200. The totals hide how bad the worst rows are. On PHerc1218 the published objective keeps 21 of its 480 estimates inside the volume and on PHerc0125 39 of 480, while the weighted sum keeps every estimate of every scroll inside. Every measurement this paper rests on is the compiled function now, and none of it is capped: the 15 scrolls of Tables III and IV, the 24 scroll run of Section V where PHerc. 0490B and the 576 of 576 live, and the exponent table, 37,152 calls in all, of which 0 met the wall clock guard. Four things are still the transcription, each named where it appears, and each for a reason that is a property of the shipped function rather than a limit of this work: the search ablation of Table VII, because the shipped function has no replaceable search; the audit of the cap and of the sampling below, because an audit of the transcription can only be measured on the transcription; the density bias on synthetic sections, because those sections are polylines of real coordinates and the shipped GridStore holds integer points; and the four scroll bench

run, which is a separate post hoc study on its own slices.

The generator makes one call disagree with the next, and that is why every row carries an interval. The generator at normalgridtools.cpp:51 is seeded from `std::random_device`, so two runs of one binary give two answers: across 4 saved slices the median distance of a run from the median of 20 runs is 0.05 to 1.31 mm, and the worst single run is 4.52 mm away. On the slice where the published version leaves the grid it does so on all 20 of 20 repeats of the compiled binary, and with the division removed not one repeat of any slice leaves it. Because of that noise every estimate below is repeated with 20 seeds per slice and reported with an interval, so that a margin can be read against the noise of the search rather than beside it. The seed argument this patch adds is what lets a row be repeated at all.

Until this revision the tables came from a transcription of the function into Python, and the ablations still do, because they turn constants the compiled function does not expose. That transcription was tied to the original where it counts for

the objective: on candidates of a real slice the score computed by the compiled function and by the transcription agree to machine precision, and the compiled function comes from a stand-in build of the source at commit 23adee047 that reproduces both values the project's own unit test asserts. The two can now be compared row by row, under a criterion fixed before the run: a row agrees when each arm's median lies inside the other arm's 95 % interval and no mark on the row changes. 28 of 30 rows agree. The column this paper claims, the weighted sum, agrees on 15 of 15: the largest move of any median is 0.18 mm, on PHerc0813 from 7.11 to 7.29 mm, the median move is 0.08 mm, and the estimate is inside the grid on all 480 runs of every scroll under both. The published column moves further, at most 3.57 mm on PHerc0125, which is what a column whose walk leaves the grid should do. No pre-registered verdict moves.

One count improved when the instrument changed, and it is named rather than absorbed. In the confirmation block the number of scrolls whose weighted sum loses to the centroid by at least 0.30 mm is 1 read by the transcription and 0 read by the shipped function: PHerc1218 has a margin of -0.32 mm by the one and -0.24 mm by the other, on either side of the line for a loss, which sits 0.30 mm below zero. That is a change of measuring instrument and not a new result, and the correction proposed here is no better for it.

2 rows disagree, PHerc0268, PHerc0813, both of the published column and neither one this paper counts as won; on both, the two medians lie inside each other's intervals and what differs is the mark for a margin whose interval includes zero, which appears on one row and disappears from the other. The rule written before the run excused that case, since it spoke of rows counted as won, and the tool that applied it does not: it tests every row and so reported these two. The tool was left stricter than its own declaration rather than edited once it was known which rows it caught.

The transcription carries one bound the C++ does not, and the tables no longer rest on it. The transcription stops at 2,000 sweeps, fixed before any result was looked at, and a walk that meets that bound is returned and scored like any other estimate, flagged in the raw files. Only the published objective ever met it: 303 of 7,200 walks of the weighted mean in the transcription's own version of the tables below, against 0 of 7,200 of the weighted sum, and 25 of 576 against 0 in the transcription's own version of the run of Section V. Every capped walk was outside the grid already, so the cap shortens a runaway and flatters the published column: rescoring those tables with the capped estimates carried 16 times further out moves 3 of 30 rows, all of them published ones and all away from the reference, by at most 0.87 mm. Verdicts changed: none (`evidence/search-cap.csv`). The uncapped run of the compiled function settles what that bound was standing in for. Of 14,400 walks, with a wall clock guard of 600 s declared before the run and applied to the two objectives alike, 0 met the guard and the longest single walk took 3.02 s. The shipped hill climb has no iteration limit and stopped by itself every time, on every runaway scroll, which is what the float candidate and float distance of `normalgridtools.cpp:110-122` predict and what the float64 transcription could not do.

B. The verdicts, applied as written

Each block carries two criteria, repair and baseline, with the threshold of 0.30 mm; both were fixed in writing before the run, in `prereg/preregistration-fifteen-references.md`, and are applied without discretion to the medians over the 20 seeds. In the primary block, removing the division gives a better median than the published function on 4 of 5 scrolls and puts the count of estimates outside the volume at zero on all of them, so repair is won. It is closer to the reference than the centroid by at least the threshold on 4 of 5 and on no scroll worse than twice the centroid, so baseline is won. The confirmation block gives 10 of 10 on repair, again with no estimate outside the volume, and 9 of 10 against the centroid, of which 2 carry a margin whose interval includes zero (PHerc0813, PHerc1545); so both criteria are won there as written, and on 2 of those 9 rows the interval does not support the margin. Against the single-seed run, the verdicts that changed with 20 seeds are: none.

The interquartile range of the per-seed medians is at most 1.03 mm, on PHerc0813. Of the 13 margins over the centroid that reach the threshold, 2 have an interval that includes zero, 0 in the primary block (none) and 2 in the confirmation block (PHerc0813, PHerc1545), and 9 are smaller than 2.205 mm, which is the distance on PHerc. 1218 between the third party's hand annotation [5] and an umbilicus derived from published sheet instance labels [6]. That derived umbilicus is a per slice centroid of the papyrus mask, so the comparison is a hand against a mask centroid and not one hand against another, and what it measures is what the centroid column already carries; Section IX gives the size of that overlap and what it costs the reading. Those margins are counted where the pre-registration counts them, because the criteria were written on point estimates and are not rewritten after the fact; the tables mark which of them the interval does not support and which fall below that distance.

One further count is descriptive and was chosen after the fact, the pre-registration having fixed the estimates outside the grid and not a distance: at 10 mm from the reference, a distance no disagreement between annotators approaches, the published function leaves 7 of 15 scrolls above that line and the weighted sum 0. It is not a criterion and nothing is declared won by it; it is reported because the worst row of the sum is 7.29 mm, so the line is cleared by a margin a reader can see rather than by a hair.

IV. Rules and readings scored after the fact

A. Two more rules, measured after the fact

Two rules the tables above do not carry will occur to a reader who knows the code: the maximum of the distance transform of the filled section, which is the rule the upstream mask based tool applies, and the plateau point nearest the centroid, the closest of the simple mask rules in the umbilicus bench [7]. Table V scores both on the slices, the heights and the control points of Tables III and IV, with the same metric, and repeats two columns of those tables so that a row reads across. Both rules are post hoc with respect to every pre-registration of this paper, and this is written after the fact, on 18 September 2026: the pre-registered baseline criterion names the centroid, it was

fixed in writing before the run, and no row of the new table changes a pre-registered verdict.

Read against the best of the three simple rules on each scroll, at the pre-registered threshold of 0.30 mm, the weighted sum is the closer on 4 of 5 scrolls of the primary block and on 7 of 10 of the confirmation block. On PHerc0813 it is ahead of the centroid by 0.30 mm, which is nothing and is counted as nothing. The maximum of the distance transform is never the closer of the two, on 0 of 15 scrolls; the plateau rule is closer on 2, and its threshold was chosen on the data of PHerc0125, PHerc0211 and PHerc0826, so on those 3 it is fitted and not tested.

The same question has been asked once more on a different sample, and the two are not mixed. That bench scores 5 mask rules on 46 heights per scroll at level 3 of the published surface predictions, heights fixed by the store and not by the extent of the reference, and both variants of the estimator were run inside that harness on its 4 scrolls. Its rows cannot be set beside the rows above: on PHerc0826 the same centroid code reads 3.12 mm on that sample and 3.51 on the slices of Table V, a difference of 0.39 mm, larger than the pre-registered threshold. What crosses from one sample to the other is that the loss falls on the same scroll and in the same direction, 4.16 mm against 2.47 for the plateau point nearest the centroid with 4 of the 5 mask rules closer than the weighted sum. It crosses too that there the published function leaves 859, 540 and 568 of its 920 estimates outside the grid on PHerc0125, PHerc0211 and PHerc0826 and none on PHerc0332, while the weighted sum leaves 0 of 3,680 over the four. That run is post hoc as well, under the same declaration, and it is the transcription's: it is not one of the two runs recomputed on the shipped function, and its slices, level and heights are its own.

B. Where it loses, where nothing can be read, and what the tests say

On PHerc0826 the centroid wins: 4.29 mm [2.90, 5.70] against 3.51 for the centroid, a margin of -0.78 mm [-2.20, 0.61], whose interval includes zero; it is called a loss because the pre-registered criterion reads the point estimate, and Section VI gives the mechanism that makes it the expected direction. Against the best of the three simple rules of Table V it loses on 2 scrolls of 15. On PHerc0826 it is 4.29 mm [2.90, 5.70] against 2.78 [1.69, 4.41] for the plateau point nearest the centroid, a gap of 1.51 mm, smaller than the 2.205 mm between the hand annotation of PHerc. 1218 and the mask centroid its instance labels give, and taken by a rule whose threshold was fitted on that scroll. On PHerc1218 it is 2.39 mm [1.62, 3.63] against 2.15 [1.64, 2.94] for the centroid, a gap of 0.24 mm, smaller than the same 2.205 mm. On PHerc0125 the constant axis at the centre of the field is closer to the reference than either estimator. That is a fact about the scroll and not about any estimator. The distance of that control is the distance from the reference to the centre of the field, and on PHerc0125 it is 1.53 mm, the smallest of the fifteen against a median of 6.73 mm. A rule that returns the field centre scores 1.53 mm there while knowing nothing about the scroll, and beating it there requires no information either. A row orders rules only where they are closer to the reference than that

control is, which the weighted sum is on all 14 of the other scrolls, where the control runs from 1.63 to 13.50 mm.

Counting scrolls is a weak use of fifteen paired measurements, and it is the use the pre-registration fixed. The tests the literature prescribes were run afterwards, on the same medians, and they disagree. A Friedman test over 15 scrolls and 4 rules rejects the null that they are interchangeable, χ^2 23.08 at p 3.9e-5, with mean ranks 1.27 for the estimator with the division removed, 2.33 for the centroid, 3.07 for the constant axis and 3.33 as published. Pairwise Wilcoxon with Holm correction then separates the repaired estimator from the centroid at p 0.0034; Nemenyi does not separate them, the gap in mean ranks being 1.07 against a critical difference of 1.21. What cannot be claimed, then, is that the weighted sum beats the centroid under every test the literature prescribes.

V. Every scroll of the challenge

Accuracy can only be measured where a reference exists. On the rest of the competition set one property can be asked without one, and it was fixed in writing before the grids were downloaded: whether the estimate lies inside the grid at all. This section is the compiled function, called through `vc_gen_umbilicus` with no cap, like everything else this paper measures. The run covers 24 scrolls, the 23 of the competition plus PHerc0332, at 24 slices each, 576 slices. As published, 350 of 576 estimates fall inside the grid, 12 scrolls keep fewer than two thirds of their slices inside, and only 3 scrolls keep all of them. With the division removed the count is 576 of 576, every slice of every one of the 24 scrolls. Both counts are the shipped function, called through `vc_gen_umbilicus` with no cap. The transcription read 351 and 576 on the same slices and seeds, so the repaired count is the same number under either instrument and the published one differs by one estimate; the difference is printed rather than reconciled. That count is over 576 *slices* at 1 seed each, the seed this run's own pre-registration fixed, and not over 576 slices times a number of seeds: the two are different claims and the smaller one is the one made here. The spread between seeds is measured separately, on 3 seeds of the weighted sum, and it is at most 1.18 % of the grid width on any scroll. It is 3 seeds and not 20 because that is the number the pre-registration of this run fixed on 2026-09-15, before the twenty seed note of Section III was written on 2026-09-16 10:59Z, and the run kept the seeds it had declared when it was repeated with the faithful sampling. Fig. 4 is one section of each of those scrolls, chosen by a rule and not by looking, with both estimates and, where one exists, the reference. On the scrolls without a reference the figure shows where the estimate falls and that it stays inside the volume; it does not show that it is right, and nothing here claims that.

VI. The cost

That the weighted sum has an interior maximum says nothing about where that maximum is. A sum rewards the side of the section that contributes more segments; a mean normalises the density away. The question was put to sections whose centre is known, with the prediction written beforehand in a note dated and marked post hoc with respect to the first



Fig. 4. One section of every scroll of the generalisation run, 24 panels sorted by scroll id. Grey: the segments of that scroll's normal grid on the slice named in the title, with the bounds of the grid as a dark grey square; each panel is drawn in the units of its own grid, whose widths run from 3,941 to 12,145 pixels, with a margin of 8 % of the width on every side. Dark grey circle: the estimate of the published estimator, the weighted mean. Blue circle: the estimate with the division removed, the weighted sum. Yellow cross: the reference umbilicus at that slice, drawn only on the 14 scrolls that have one, published by the challenge on PHerc. 0125, PHerc. 0211, PHerc. 0332, PHerc. 0826 and annotated by a third party on PHerc. 0191, PHerc. 0257, PHerc. 0268, PHerc. 0358, PHerc. 0800, PHerc. 0813, PHerc. 1203, PHerc. 1218, PHerc. 1447, PHerc. 1545; the reference was used only to measure and never to choose anything, and the 10 scrolls PHerc. 0175A, PHerc. 0175B, PHerc. 0306B, PHerc. 0343, PHerc. 0483A, PHerc. 0483B, PHerc. 0490A, PHerc. 0490B, PHerc. 0846A, PHerc. 0846B have none. Dark grey arrow: the published estimate lies beyond the window, in the direction the arrow leaves the blue circle, at the labelled distance between the two estimates, in millimetres where the voxel size of the scan is on record and in grid widths otherwise; this happens on 11 of the 24 panels. On the slices drawn, the published estimate lies outside its grid on 11 panels, PHerc. 0125, PHerc. 0211, PHerc. 0306B, PHerc. 0358, PHerc. 0483A, PHerc. 0490B, PHerc. 0846A, PHerc. 0846B, PHerc. 1203, PHerc. 1218, PHerc. 1545, and the estimate with the division removed lies inside on every panel. No walk in this run was stopped by a cap: the estimates are those of the compiled function, which has no iteration limit, and every walk stopped by itself. Slice rule: of the 24 slices the run sampled on each scroll, evenly spaced along z , the panel shows the twelfth in order of z , the lower of the two that straddle the middle of the sampled band; no slice was chosen by looking at it. Over the whole run, 576 of 576 estimates fall inside the volume with the division removed, against 350 of 576 as published. Drawn by tools/figure_umbilicus_gallery.py from evidence/cpp-positions-24.csv, the cached grids of the run and the reference files; no estimate was recomputed.

version of this measurement (prereg/note-density-bias.md). On a full circle whose lower half is thinned to one segment in ten, the maximum of the sum would move towards the dense side by 50 to 250 grid units and the maximum of the mean by less than 10.

Table VI answers. On the uniform circle the maximum of the sum sits 47 units from the true centre and that of the mean 0; with the lower half thinned ten times the maximum of the sum moves +242 units along x towards the dense side, 2.29 mm, and the maximum of the mean +0. The prediction for the sum held and the prediction for the mean held. The

bias grows with the ratio and is largest on the half circle, where one side has no segments at all: 261 units from the centre for the sum, 0 for the mean. The full estimate follows the maximum in every row. On these sections the published version is the more accurate one, 0.004 mm from the true centre on concentric circles against 0.430 mm for the sum (evidence/synthetic-sections-refit.csv). The mean is exact wherever it has a maximum to find, and the sum pays for its guaranteed maximum with a bias towards density of 0.44 to 2.44 mm on these sections. Those millimetres are distances from a centre known by construction, and they are

TABLE VI

Synthetic sections with a centre known by construction, 8000 by 8000 grid units, radii 300 to 3500. Position of the maximum of each objective on a lattice of step 4, and of the full estimate with the search of `normalgridtools.cpp:78-142`, medians over 20 seeds, in grid units (1 unit = 9.362 μm). Error: distance from the true centre. Δx : offset along x , positive towards the side that holds more segments. The density rows keep every segment with x above the centre and one in 2, 5 and 10 of those below it; the half circle keeps the half with y above the centre, so its offset is along y and its error column carries it. Generated from `evidence/synthetic-density.csv`.

section	sum, maximum error		mean, maximum Δx		estimate sum mean	
circle 1:1	47	-2	0	+0	46	0
ellipse 1:4	125	-36	6	+0	126	6
half circle	261	-12	0	+0	259	0
density 1:2	202	+200	0	+0	202	0
density 1:5	229	+226	0	+0	230	0
density 1:10	244	+242	0	+0	244	0

not a term to be added to the distances of Section III: those are measured from the estimator's own output against a hand annotation, so whatever bias a real section induces is already inside them and cannot be counted twice. The widest of the synthetic biases comes from a section with one side empty, which no slice of a real prediction is.

The same measurement was made on the real predictions, first on the five primary scrolls and then on all fifteen. The segments of the 24 slices of each scroll were counted in eight angular sectors around the reference axis, which serves only as the origin of the sectors, and the sector ratio of a scroll is the count of its most populated sector over its least populated one, median over the slices (`evidence/density-asymmetry-fifteen.csv`). On the primary block the ratio orders the scrolls exactly as the estimator's distance from the reference does, Spearman +1.00 over 5 scrolls, and PHerc0826, the one primary scroll where the centroid wins, has the highest ratio of the fifteen, 6.38 against a median of 3.59. That ordering does not survive the fifteen: over all of them the Spearman correlation between the ratio and the distance from the reference is +0.33 (p 0.23) and between the ratio and the margin over the centroid -0.35 (p 0.20). The two confirmation scrolls where the centroid wins or nearly wins, PHerc1218 and PHerc0813, are not the one-sided ones by this measure: their ratios are 4.08 and 3.59, ranks 10 and 8 of 15 counted from the least one-sided, so the prediction that they would be asymmetric is not confirmed, and no second measure of one-sidedness is offered here to rescue it. The density bias is a known limit of the change, and on the real scrolls it explains the one primary loss and not the ranking: the one line trades an unbounded failure for a bounded bias towards the side of the section the prediction happens to have found, and on a very one-sided prediction that bias can exceed the distance of the centroid baseline.

VII. The objective, not the search

A first assumption was that the search was at fault as well, and it was rewritten: double precision throughout, the walk confined to the volume, a step that halves when none of the eight probes improves instead of chaining moves inside the

3 by 3 loop, and one objective instead of two. Table VII measures the four combinations on the same slices and seeds as Section III. With the weighted mean, the rewritten search keeps 7,200 of 7,200 estimates inside the grid against 4,484 for the published walk, because it is clipped there, and still lands 10 mm or more from the reference on 8 of 15 scrolls, at worst 39.97 mm, much as the published walk does (7 scrolls, at worst 133.02 mm). With the weighted sum neither search fails on any scroll, and their worst rows are 7.11 and 6.50 mm. The rewritten search beats the centroid by the pre-registered margin on 4 of 15 scrolls with the mean and on 13 with the sum, against 13 for the one line; it is better than the one line in the median on 11 of 15 scrolls with the sum and on 1 with the mean. A well posed objective makes even a loose search converge, and the estimates outside the volume disappear without a clamp because the score falls away in every direction, so no step outwards keeps improving it.

Column (d) is closer on most of the scrolls, and the size of that difference is what decides the patch. With the sum, the rewritten search is closer on 11 of 15 scrolls, further on 3 and level on 1, by a median of 0.06 mm, gaining at most 0.61 mm and losing at most 0.51; the medians over the fifteen are 1.84 mm against 1.87. Only 1 of the 15 reaches the pre-registered 0.30 mm in its favour (PHerc0813) and 1 reaches it against (PHerc0191), so by the rule the rest of this paper decides on, the two are not separated. Column (c) settles what the rewrite repairs on its own: with the mean it still lands 10 mm or more from the reference on 8 scrolls and beats the centroid on 4 of 15 against 13 for the line, so the search was not the defect. That is the argument for one line rather than for a rewrite: the rewrite is four changes to a search and a new one to review, it buys a median of 0.06 mm where the objective is right, and it repairs nothing where it is wrong.

VIII. The patch

The commit proposed upstream, 7a4129a1d on the fork, touches 3 files and carries the tree 96a769b3; the tree is what every measurement of this paper was made on, and the tree is given beside the commit because a commit hash changes whenever the message is reworded while the code does not. Line 116 returns `score` instead of `score/wsum` and the accumulator `wsum` goes. The comment above the return states the trade of Section VI rather than presenting the change as a repair without cost. It says that the sum favours candidates near many segments as well as candidates the normals point at; that on synthetic sections with a known centre the mean is exact and the sum is biased towards the denser side; and that on the fifteen real scrolls the failure persists with the walk clipped to the grid. Two alternatives are not in the patch and were not measured: the mean multiplied by a coverage term that counts the normals, and a weight that falls faster than $1/D$ so that the mean itself decays. Each is a new objective, and Section IX says why a new objective is not offered from this bench before being measured on it.

The three changes in the patch answer three different questions and should be read separately: the line changes the objective and carries the measured claim of this paper, the seed argument makes a run repeatable and claims nothing about the

TABLE VII

The four cells of the ablation on the fifteen scrolls, median distance from the reference in millimetres, 20 seeds per slice, with the 95 % bootstrap interval of Section III. (a) as published: weighted mean, the search of `normalgridtools.cpp:78-142`. (b) the weighted sum, search untouched: the one line. (c) the search rewritten (double precision, walk clipped to the grid, eight probes from a fixed centre and a step that halves when none improves, one objective for seed and refinement), weighted mean. (d) the rewritten search, weighted sum. This table is the transcription's, columns (a) and (b) included. An ablation is measured on the instrument that can ablate the thing, and the shipped function has no replaceable search: columns (c) and (d) replace it, which is a property of `align_and_extract_umbilicus` and not a limitation of this work. Columns (a) and (b) are therefore the same quantity as Tables III and IV read by the other instrument, and they differ on 26 of 30 cells, by at most 3.57 mm. Before the tables were recomputed on the shipped function they matched row for row, which is what the two columns were built to do. Generated from `evidence/ablation-search.csv`.

scroll	(a) weighted mean, as published	(b) weighted sum, the one line	(c) rewritten search, mean	(d) rewritten search, sum
PHerc0125	133.02 [107.99, 190.24]	1.59 [1.15, 2.17]	39.96 [39.50, 40.60]	1.64 [1.18, 2.21]
PHerc0211	63.81 [45.21, 95.60]	1.47 [0.98, 2.38]	37.97 [33.43, 40.24]	1.32 [0.85, 1.97]
PHerc0332	0.35 [0.28, 0.41]	0.36 [0.30, 0.42]	0.35 [0.28, 0.41]	0.36 [0.30, 0.42]
PHerc0826	53.21 [40.75, 66.12]	4.27 [3.03, 5.44]	39.97 [37.27, 41.35]	4.30 [2.81, 5.37]
PHercParis4	4.25 [1.47, 26.57]	1.13 [0.83, 1.48]	3.12 [1.37, 11.04]	1.07 [0.81, 1.41]
PHerc0191	6.28 [4.39, 8.50]	4.15 [2.64, 6.03]	6.22 [4.56, 8.23]	4.66 [3.21, 6.02]
PHerc0257	8.37 [2.75, 31.17]	1.87 [1.26, 2.72]	5.43 [2.17, 30.40]	1.84 [1.28, 2.73]
PHerc0268	8.27 [5.63, 11.05]	4.32 [2.69, 5.87]	8.18 [5.24, 10.90]	4.30 [2.79, 5.95]
PHerc0358	8.46 [3.92, 97.85]	2.89 [1.74, 3.86]	10.38 [3.70, 37.37]	2.75 [1.63, 3.87]
PHerc0800	5.29 [3.49, 36.18]	2.68 [1.73, 3.83]	4.80 [3.30, 33.00]	2.66 [1.63, 3.70]
PHerc0813	10.49 [7.10, 12.45]	7.11 [4.42, 9.33]	10.26 [7.62, 12.29]	6.50 [4.31, 8.69]
PHerc1203	4.80 [1.46, 34.91]	1.44 [0.98, 1.77]	2.85 [1.40, 16.83]	1.38 [1.01, 1.75]
PHerc1218	104.09 [82.99, 150.90]	2.47 [1.66, 3.89]	33.27 [29.88, 35.68]	2.24 [1.60, 3.62]
PHerc1447	38.57 [28.75, 55.24]	1.76 [1.15, 2.56]	33.20 [25.73, 38.13]	1.69 [1.17, 2.51]
PHerc1545	49.55 [17.68, 68.47]	1.87 [1.24, 3.38]	32.66 [9.14, 40.16]	1.80 [1.25, 3.19]

objective, and the test bounds the bias the line introduces rather than showing the line to be right.

The generator that draws the 10,000 sampled segments and the 1,000 candidate points was a static `std::mt19937` seeded from `std::random_device`, so no run could be repeated. The signature gains one argument with a default:

```
cv::Vec2f align_and_extract_umbilicus(
    const GridStore& grid_store,
    std::optional<std::uint32_t> seed = std::nullopt);
```

With no seed the behaviour is exactly as before, the one static generator seeded once from the hardware; with a seed a fresh generator is seeded with that value and the run repeats exactly. The function has no other caller in the tree, so the change is source compatible.

The regression test adds 4 cases to the function's own test file, `test_normalgridtools.cpp`. One asserts that two calls with the same seed return the same point. Three more are on `GridStore` sections built as in Table VI, concentric circles, a half circle and circles thinned 1:10 on one side, estimated with a fixed seed and asserted inside the grid and within 1.0, 5.0 and 4.5 mm of the known centre; the thinned section also asserts that the estimate leans towards the dense side. The tolerances are derived from the table, not from a run: each is at least 1.56 times the p90 over 20 seeds of the estimate's error on that section (0.61, 3.20 and 2.71 mm), so that no seed fails them by statistics alone. The project's suite runs 20.50 per cent of the lines of this function and reaches neither the objective nor the search, so nothing in it could have disagreed with the division.

The test was built from the commit's own sources: the target needs OpenCV core and `imgproc` and not Qt, which the test directory's CMake demands for the viewer tests and not for this one. The flags are not the project's CMake flags and the OpenCV is the distribution's rather than the one of the project's image, and neither touches the arithmetic under test. All 15 cases of the file pass. The 4 new ones put the estimate 0.23 mm from the known centre on the

concentric circles against a tolerance of 1.0, 2.13 mm on the half circle against 5.0, and 2.02 mm on the thinned circles against 4.5, that estimate at x 4206 and so on the dense side of the centre as Section VI predicts. Every estimate is inside the grid, the repeated seed gives the same point to the bit, and a second run in a fresh process gave the same 5 estimates (`evidence/regression-test-run.csv`). What the test protects is said in its own comment: it cannot tell the sum from the mean, because on these sections the mean is exact; it bounds the sum's known bias so that it cannot grow unnoticed, and it checks that the walk settles inside the grid.

A. The exponent of the weight

The exponent of $1/\max(100, d)$ came with the function and the patch does not touch it, which is a choice to be measured rather than assumed. The estimator was rerun over the 15 scrolls of Section III, the same 24 slices and 20 seeds, with the exponent set to 0.5, 1, 1.5, 2 and nothing else changed; at $p = 1$ the code takes the original expression and reproduces this paper's own estimates exactly, 0 differences over 0, so Table VIII can be read beside every other number here.

Robustness does not choose an exponent and accuracy does. Every exponent tried keeps all 7,200 estimates inside the grid against 2,727 outside for the published mean, which is the algebra of Section II again: the repair comes from not dividing, not from the value of p . Measured on 3 sections and 5 exponents up to $p = 4$, at 4 distances a decade apart out to 10^9 units, the departure of the mean from $\lambda_{\max}$ divides by 8.7 to 10.5 a decade, which is $O(1/D)$, and the sum divides by 10^p : the bar fixed before that run is met on all 15 of 15 combinations (Fig. 2). The median over the fifteen scrolls, on the other hand, is 3.38, 1.87, 2.44 and 4.99 mm at 0.5, 1, 1.5, 2, lowest at 1 and interior to the range tried rather than at its edge, while the declared cost grows with the exponent on both fronts at once, the density bias and the spread between seeds.

TABLE VIII

The exponent of the weight on the 15 scrolls with a manual reference, 24 slices and 20 seeds each, everything but the exponent as in Table III. Outside: estimates falling outside the grid. Median: the median over the 15 scrolls of the per scroll median distance from the reference, with a 95 % percentile bootstrap interval (1,000 resamples, generator seed 20260916). The unit resampled there is the scroll and not the control point, because the quantity is a median over scrolls; the arithmetic is that of Tables III and IV and only the unit differs. That median and its interval run over the two blocks together, as this row did before the interval was added. Pooling is admissible here because no pre-registered verdict of this paper is taken from a pooled sample. Beats: scrolls where the margin over the centroid reaches the pre-registered 0.30 mm. Spread: the median spread between seeds. Bias: the density bias on the section thinned ten times, as in Table VI. The last row is the published weighted mean on the same runs. Every row here is the compiled function, called through `vc_gen_umbilicus`, but three of the five rows are compiled variants of an objective upstream does not ship: the weight at `normalgridtools.cpp:112` is written `1.0f / std::max(100.0f, dist)` with no exponent, so p of 0.5, 1.5 and 2 are one line edits of it, built by a substitution anchored on that exact line which fails the build rather than produce a binary nobody can characterise. So on those three rows the compiled figure carries no more authority than the transcription's did; what it buys is that p equal to 1, which is the shipped expression itself, and the control row, which is the shipped weighted mean, now agree with Tables III and IV to the digit instead of reading 8.46 mm against 9.38 mm for one quantity on one page. The bias column is the one thing in this table still measured by the transcription: its sections are polylines of real coordinates and the shipped `GridStore` holds integer points, so the shipped function cannot be given them without rounding them first. Generated from `evidence/cpp-exponent-fifteen-k20.csv`, `evidence/cpp-exponent-aggregate.csv` and their companions.

p	outside	median mm	beats	spread mm	bias mm
0.5	0 of 7,200	3.38 [2.09, 5.54]	10 of 15	0.34	1.80
1.0	0 of 7,200	1.87 [1.59, 2.76]	12 of 15	0.57	2.29
1.5	0 of 7,200	2.44 [2.23, 3.08]	12 of 15	1.04	4.48
2.0	0 of 7,200	4.99 [3.50, 6.01]	5 of 15	3.20	5.62
mean	2727 of 7,200	9.38 [5.16, 49.49]	3 of 15	–	0.00

Those medians are not separated by their intervals. At 1 the interval is [1.59, 2.76] mm, and of the exponents tried only 2 lies clear of it; 0.5 and 1.5 overlap it, so the aggregate alone does not put 1 ahead of them. It also throws away the pairing, and the pairing is what the rest of this paper decides on. Scroll by scroll, at the pre-registered 0.30 mm, no exponent tried is better than 1 on more than 1 of the 15 scrolls, the one being 1.5, while each is worse on 8 to 13. That is the comparison the exponent is left at 1 on, and it is a reason not to move rather than a demonstration that 1 is right.

One reading of that table can be turned against the choice. The smallest exponent, 0.5, passes three of the four bars written before the run: it is as robust, its spread between seeds is smaller, 0.34 against 0.57 mm, and its density bias is smaller too, 1.80 against 2.29 mm on the thinned section. It fails on the fourth, accuracy, which asked for a gain of 0.30 mm on at least 8 of the 15 scrolls and a loss of that size on none: 0.5 is better on 0 scrolls and worse on 10. A reader who puts the density bias ahead of accuracy would prefer it, and the rule that puts accuracy first was written before any of these numbers were seen and is applied as written. An argument for 0.5 would have to declare that order first and make the choice again on a run of its own, not reread this table.

IX. What this does not establish

It does not establish that the weighted sum is the right objective. On complete concentric sections the mean is the

more accurate one, by the margin of Section VI, and a reader working on such sections should keep it; the division is wrong on the partial, one-sided predictions this toolchain actually processes, where it lets the search leave the volume, and the claim is made in that narrower form.

It does not establish accuracy, and the error of the references is not known. The distances of Section III are distances from hand annotations whose own error is not known, and on today's public data it is not measurable on any scroll: the 5 umbilici the challenge publishes and the 10 clicked by the third party fall on 15 different scrolls, one hand each, and the two sets do not meet. What PHerc. 1218 carries beside its hand annotation is not a second hand. It is an umbilicus derived from published sheet instance labels [6], and the script that derives it takes the centroid of the papyrus pixels of each slice, reading the labels only as a mask and never which sheet is which, so the second reference is the construction this paper scores as its centroid baseline. The 2.003 to 2.205 mm that separate the two on that scroll are a hand against a mask centroid, 9 of the 13 margins won are below that distance, and what they are below is the baseline's own error rather than the error of a hand.

How far apart two reference producing methods sit is a different statement, and it can be made on more than one scroll. The same derivation runs from the masked scan and the surface prediction the challenge publishes, so it was repeated on every scroll of Section III: 15 pairs, a hand against a mask centroid, with the distance running from 0.766 mm on PHerc0332 to 10.748 mm on PHerc0268, median 3.660 mm, quartiles 2.681 and 5.942, a ratio of 14.0 between the largest and the smallest. PHerc. 1218, the one scroll the 2.205 mm was measured on, is the second smallest of the 15, and 13 of the 15 lie above 2.205 mm, so that number is not a scale for the others and is not used as one here. Over those 15 scrolls the distance tracks the centroid column of Tables III and IV with a Pearson correlation of 0.9954 and a median difference of 0.19 mm, which is the overlap the dagger of Table III inherits: the two are close to the same quantity, and no criterion of this paper rests on either. Read from `evidence/second-reference.csv`, written by `tools/second_reference.py`.

One thing the measurement suggests and the patch does not do: the far field bound of equation (1) is a byproduct of the samples the function already draws, so the code could warn when that bound exceeds the score it is about to return. The two quantities are not the same, and the difference is the reason the warning is not offered here. Table II compares the bound with the weighted mean at the reference control point, which the code does not have, and on that comparison it flags 134 of the 135 walks that left the grid at a cost of 58 false alarms out of 192. Compared instead with the refined score, the only score the function holds when it returns, the same test fires on 21 slices of 360, 4 of them on a run that stayed inside, and it misses 119 of the 136 runs that left on that single seed, against the 135 slices of Table II counted by majority of the 20: sensitivity 0.12 against 0.99. Once the walk has reached the far field its own score has risen to the bound, so the comparison that was true where the search started is false where it stops. The useful form of the diagnostic is therefore

not the one measured here but the one made where the search starts: the seed of the refinement, the best of the 1,000 random candidates of `normalgridtools.cpp:78-88`, is a point the function already holds, and the weighted mean there is one more call. That is the thing to measure, and it is not a result.

Three things in the same function are real and are not in the patch, each because it is a new objective or a new search that would have to be measured on this bench before being offered. The walk is unbounded: with the sum the score falls away in every direction and on these runs the walk settles inside the grid every time, but that is measured, not guaranteed. There is no consensus set, despite the comment that announces one: every sampled segment contributes its squared cosine with no threshold and no outlier rejection, so a segment whose normal is wrong is averaged in rather than discarded. And the seed score of `normalgridtools.cpp:78-88` and the refinement score of `normalgridtools.cpp:98-117` are two different functions, so the search can hand the refinement a starting point its own objective would never have chosen.

X. Conclusion

What the measurements support is narrow and can be stated in one line: the normalisation in the refinement score produces a far field attractor for this estimator, and removing it eliminates the failure mode on every scroll measured, at the price of a density bias that has been measured too. The attractor is not a manner of speaking. Dividing by the sum of the weights leaves a quantity whose limit far from the section is the largest eigenvalue of the covariance of the normals, which on a partial prediction can exceed the score on the axis. Where it does, the hill climb is right to walk out, and it walks out on 135 of 360 slices, 134 of those departures satisfying that condition, the remaining one, on PHerc. 0257 z 8479, sitting at the majority line under the shipped walk. Without the division the estimate lands inside the volume on 576 of 576 slices of 24 scrolls against 350 as published, and on the 15 scrolls with a reference it lands 10 mm or more from it on 0 against 7, a count chosen after the fact and cleared by a visible margin, the worst row of the sum being 7.29 mm. The price is a bias, not a failure: the sum leans towards the denser side of a section by 0.44 to 2.44 mm on synthetic sections whose centre is known by construction, and is therefore not a term to add to the distances above. On the real scrolls the one-sidedness of the segments explains the one loss in the primary block and does not order the fifteen.

What the measurements do not support is that the weighted sum is the correct geometric objective for an umbilicus. On complete concentric sections the mean is the more accurate of the two, and a reader working on such sections should keep it. The accuracy margins over the centroid cannot be set against the error of a hand annotation, because on today's public data that error is not measurable on any scroll: the one second reference, on PHerc. 1218, is a mask centroid and not a second hand, and Section IX says what a comparison against it is worth. And the alternatives that would have to be measured before any of them could be called correct, a mean with a coverage term, a faster decay, the comparison at the seed point of Section IX, have not been measured here. The

claim is the one the evidence carries, that an objective with no interior maximum has been replaced by one that has it, at a known cost, and the change offered is therefore the line, with a seed argument that makes a run repeatable and a test that keeps the bias bounded.

Code and data availability

Everything behind this paper is in `research/00001/` of the public repository [8]: the two pre-registrations, in the original and in English, the evidence files every number is read from, the tools that write them and the sources of this paper. The patch is commit `7a4129a1d`, tree `96a769b3`, on the branch `normalgrid-umbilicus-score-not-normalised` of the author's fork of `villa`. Code is MIT licensed, the paper and its figures are CC BY 4.0, and everything was computed on CPU.

Acknowledgements

`volume-cartographer` is the Vesuvius Challenge's, and so are the five reference umbilici of the primary block, one of them annotated by David Josey. The ten references of the confirmation block are the third party's `herculaneum-umbilici` [5], MIT; the second reference of PHerc. 1218 comes from `vesuvius-sheet-tools` [6], MIT, and is derived from a published set of sheet instance labels as a per slice centroid of the papyrus mask, not clicked. `Umbilicus_Maker` [2] is the manual tool an automatic estimate is meant to spare.

About the author

Giovanni Pellerano is a computer engineer and whistleblowing hacktivist. Co-author and Project Lead of the free and open-source software, `GlobalLeaks`, he is responsible for software development and for the support of the community that has grown around it worldwide. He is a digital rights advocate currently learning more about mainstream generative AI development and how to resist against their various and profound harms.

He had not heard of the Vesuvius Challenge before Saturday 5 September 2026. Everything reported in this series was done after that date, which is stated because it bounds what the work can be assumed to rest on: no prior familiarity with the data, the tools, the community or the literature, just a curious and hyperfocused mind joining this new journey.

Declaration of generative AI in the research and writing process

During the preparation of this work the author used Claude Opus 5 (Anthropic) and, for the revision of 16 September 2026, Claude Fable 5.1 (Anthropic) in order to carry out the full research and to write the article: the model read the data, wrote and ran the tools, performed the measurements, and drafted the text. The author set the direction, supplied the domain judgement, challenged the conclusions, reviewed and edited the content, decided what was published, and takes full responsibility for the content of the publication.

The failure this arrangement is prone to is asserting a conclusion that no measurement supports. It happened repeatedly in the work this paper is cut from. The conclusions that did not survive measurement are not in this paper; the longer paper of the series keeps them and their retractions in its text, so a reader can see both what was claimed and what the measurement did to it.

What keeps the arrangement checkable is in the folder and not in this paragraph. The criterion of a study is registered before the run, with the day and the hour it was written, and is not revised once the results are in: the pre-registrations this article rests on are in `src/prereg/`, with their digests. Every number printed here expands a macro computed from a file in `src/evidence/`, and each of those files is written by a tool in `src/tools/`: none is typed by hand, by the author or by the model. A reader who doubts a number is not asked to trust the arrangement: the file, the tool and the command that produced it are in the same folder.

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